EMCH 574 – Vibration and Waves
Homework 02

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Assignment

- G signifies students that take this class for Graduate Credit; they must do all the items
- UG students are not required to do the G work, but they may do it for bonus points
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional; they will earn bonus points.

Notes

- Show your work to obtain partial credit if appropriate
- Always display input data!
- Partial credit might be given based on work done and MATLAB codes if they are attached; without them, no partial credit can be given.
- Display numerical results to three significant digits (four if the first digit is 1).
- Use SI system of measurement reference
- Scale units using appropriate prefixes (e.g., nano, micro, milli, kilo, mega, giga...) reference
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional. They may be solved for extra credit
- Normalized plots are plots with the y-axis scaled between -1 and +1. A plotted variable is normalized by dividing it by its maximum absolute value

Notations and Abbreviations

- BVP = boundary value problem
- EOM = equation of motion
- FBD = free body diagram
- FRF = frequency response function
- IVP = initial value problem
- LHS = left hand side
- NLM = Newton's law of motion
- ODE = ordinary differential equation
- PDE = partial differential equation
- RHS = right hand side

1 Waves

A.1 Wave Fundamentals

(1) List at least five examples of waves met in everyday life

(2) Explain in your own words the difference between disturbance waves and continuous waves

(3) Explain using text and sketches the definition of wavespeed for a disturbance wave

(4) Give at least three different equivalent expressions for a disturbance wave

(5) A car pileup takes place on an interstate and propagates as a wave for several miles. What type of wave is this? Choose one of the following:

- **(a)** continuous wave
- **(b)** disturbance wave

(6) Explain using text and sketches the definition of wavespeed for a continuous harmonic wave

(7) Explain using text and sketches the definition of wavelength and wavenumber for a continuous harmonic wave

(8) Give at least three different equivalent expressions for a continuous harmonic wave

(9) What do waves carry? Choose one of the following:

- **(a)** matter
- **(b)** energy
- **(c)** both
- **(d)** none of the above

(10) Graduate Work: write about the particle-wave duality of light rays

A.2 Simple Wave Calculations

Take electromagnetic wavespeed $c_{\text{light}} = 300,000\,\text{km/s}$, sound wavespeed $c_{\text{sound}} = 340\,\text{m/s}$. Do the following quick calculations:

(1) Calculate the frequency of visible light with typical wavelength $\lambda = 510\,\text{nm}$

(2) Calculate the Hz frequency f of an AM radio wave with wavelength $\lambda = 550\,\text{m}$.

(3) Infrared radiation has a typical wavelength $\lambda = 10\,\mu\text{m}$. What is the typical frequency of infrared radiation?

(4) Calculate the wavelength λ of an FM radio wave with frequency of 89 MHz.

(5) The lightning flash is observed at 8:23:$32\,\text{pm}$ whereas the thunder is heard at 8:23:$36\,\text{pm}$. How far is the lightning strike? Give your answer in km and miles.

(6) One of Verizon's 5G mobile signals has about 29 GHz. What is the corresponding wavelength?

(7) Microwave oven typically operates at 2.5 GHz:

- **(a)** what is the wavelength λ?
- **(b)** what is the maximum permissible hole size D in the oven's door mesh considering that size of the largest hole in a Faraday cage must be at most 1/2 of the wavelength?
- **(c)** measure approximately the hole size in a microwave oven door mesh and discuss your results. You may search the web for various opinions on this matter.

(8) Graduate Work: given that the diameter of an electron is estimated to be $D = 2\,\text{pm}$, do the following:

- **(a)** calculate the frequency of the electromagnetic wave that has a wavelength equal to the electron diameter
- **(b)** in what spectrum range would this electromagnetic wave be? (options: infrared, visible, ultraviolet, X-ray, gamma ray)

A.3 Calculate Car Motion On Wavy Road

Consider a car traveling on a wavy road. The road surface undulates with an amplitude $a = 2.5\,\mathrm{in}$ and a wavelength $\lambda = 105\,\mathrm{ft}$ that follows a cosine shape. The car is modeled as a spring-mass-damper (Figure 1) with natural frequency $f_n = 0.95\,\mathrm{Hz}$ and damping ratio $\zeta = 22\%$. The maximum car speed is $v_{\max} = 125\,\mathrm{mph}$.

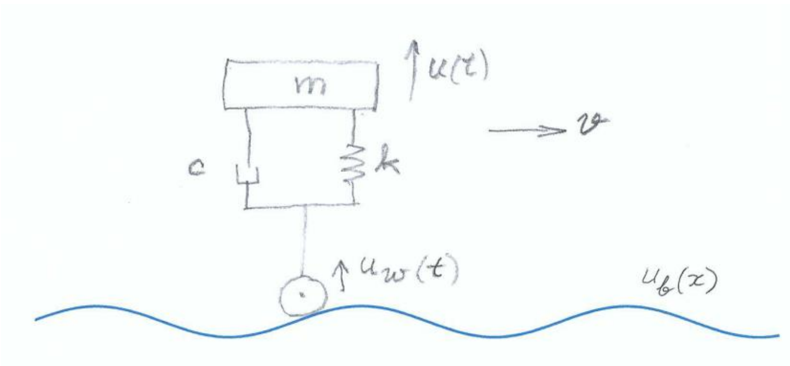

Figure 1 Car riding on a wavy road

Do the following:

(1) Display input data

(2) Calculate wavy road wavenumber γ

(3) Plot the wavy road profile $u_b(x)$ for $N_\lambda = 2$ wavelengths

(4) calculate wavy-road frequency $f_{\max}$ for a maximum car speed of $v_{\max}$ and define the frequency range $f = \mathrm{linspace}(0, f_{\max}, N_f)$ with $N_f = 2000$

(5) plot $|\mathrm{FRF}(f; f_n)|$ for. Use datatips on the plot to locate the peak response and the matched-frequency response.

(6) Calculate the matched-frequency response value $|\mathrm{FRF}_n|$. Compare with the matched-frequency readings at item (5)

(7) Calculate the car speed v_n needed to make wavy-road frequency f match the car natural frequency f_n

(8) Use the conversion $f = v/\lambda$ to plot $|\mathrm{FRF}(v; f_n)|$. Use datatips on the plot to find:

- **(a)** v_n and $|\mathrm{FRF}_n|$ and discuss them in comparison with results of items (6), (7)
- **(b)** v_{cr} and $|FRF_{\max}|$
- **(c)** the limit speed v_{limit} for which the motion amplification is limited to $|\mathrm{FRF}_{\mathrm{lim}}| = 1.5$

(9) Graduate work: Find the road wavelength λ_{new} that will make $v_n = 35\,\mathrm{mph}$

(10) Graduate work: Repeat items (5) through (8) for the case of a car with worn-out shocks that can only give $\zeta = 10\%$ and comment on the comparison between the two cases

A.4 Standing Waves

(1) Follow the notes to develop the standing-wave concept:

- **(a)** write the complex-exponential expressions for two identical harmonic waves of amplitude 1/2 propagating in opposite directions
- **(b)** add the two waves and write the resulting complex- exponential expression
- **(c)** use Euler's formula to arrive at a trigonometric expression

(2) Sketch a standing wave and identify the nodes and anti-nodes (bows)

(3) Assume wavespeed $c = 340\,\mathrm{m/s}$ and frequency $f = 2.9\,\mathrm{kHz}$. Calculate:

- **(a)** rad/sec frequency ω
- **(b)** wavenumber γ
- **(c)** wavelength λ
- **(d)** distance Δx between standing wave nodes

(4) Graduate Work: explain intuitively the analogy between standing waves and the vibration of a taught string

A.5 Musical Notes

(1) Find the Hz frequency of note C_5

(2) Find what musical note corresponds to $246.94\,\text{Hz}$

(3) What is the lowest frequency on the scale of equal temperament and what musical note does it correspond to?

(4) What is the commonly accepted frequency range of human hearing?

(5) What is the highest frequency on the scale of equal temperament and what musical note does it correspond to?

(6) Graduate Work: explain the scale of equal temperament:

- **(a)** what is the organizing principle of the scale of equal temperament
- **(b)** how many equal-temperament entries are in an octave?
- **(c)** what is the ratio between two adjacent entries on the scale of equal temperament?

A.6 Musical Instruments

(1) List at least three methods to produce sound in a string instrument

(2) List the two different methods to produce a sound in wind instruments

(3) List string instruments that use a hammer hit to produce sound

(4) List the three major categories of musical instruments. In each category, do the following:

- **(a)** describe how the instrument works
- **(b)** explain what physical principles are used to produce sound

(5) List at least five common string instruments

2 Fluid Waves and Vibration

B.1 Analyze Wave Equation In Fluids

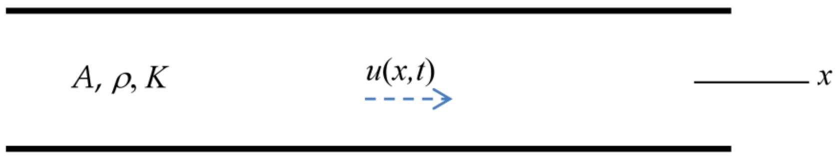

Figure 2: fluid waves in an infinite tube

Assume an infinite tube of cross section A containing fluid of density ρ and bulk modulus K (Figure 2). Develop the wave equation in fluids. Do the following:

(1) assume that the fluid in the tube undergoes time-varying axial displacement $u(x,t)$ under the influence of a pressure disturbance $p(x,t)$

(2) Consider an infinitesimal element dx and draw the displacements and pressures on both of its sides

(3) draw FBD

(4) apply NLM and deduce relationship between pressure p, density ρ and acceleration $\ddot{u}$

(5) apply fluid-elasticity equation and express the pressure $p(x,t)$ in terms of displacement $u(x,t)$ and bulk modulus K

(6) substitute (5) into (4) deduce the relationship between second derivative wrt space u'' and second derivative wrt time $\ddot{u}$

(7) define the wavespeed c

(8) substitute (7) into (6) and deduce the wave equation

B.2 Analyze And Find Solution Of Wave Equation

Recall the results of Problem B.1 and do the following:

(1) recall the wave equation

(2) assume separation of time and space behaviors into $U(x)$ and $T(t)$ and write $u(x,t)$ as a function of $U(x)$ and $T(t)$

(3) do the appropriate differentiations wrt to space and time

(4) assume vibratory time behavior with angular frequency ω using a complex exponential formula for $T(t)$

(5) substitute (4) into (3) and write expressions for u'' and $\ddot{u}$

(6) substitute (5) into the wave equation (1) and express it only in terms of $U(x)$, c, ω

(7) define wavenumber γ

(8) substitute (7) into (6) and obtain a linear ordinary differential equation (ODE) in terms of $U(x)$, $U''(x)$, and γ

(9) assume exponential solution of the form $U(x) = e^{sx}$

(10) substitute (9) into (8) and obtain the characteristic equation

(11) solve the characteristic equation (10) and obtain the roots s_1, s_2

(12) use (11) into (9) and write the complete solution in terms of constants A_1, A_2 and complex exponentials of s_1, s_2

(13) add the vibratory time behavior from (4) to obtain the solution in terms of complex exponentials depending on both γ and ω

(14) use Euler's formulae to rewrite (13) in terms of new constants C_1, C_2 and trigonometric functions depending on γ and ω and

B.3 Analyze Air Vibration In A Tube Closed At Both Ends

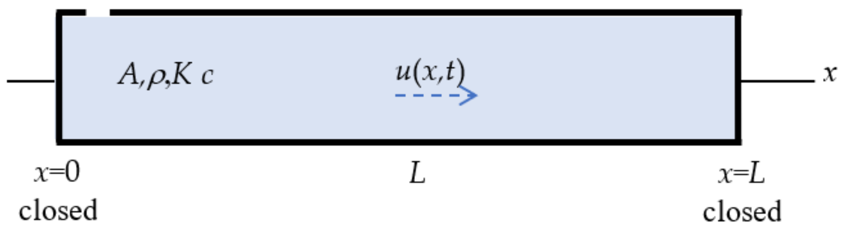

Figure 3: air vibration in a tube of finite length L closed at both ends

Assume a tube of finite length L and cross section A containing fluid of density ρ, bulk modulus K, and wavespeed c (Figure 3). The tube is closed at both ends. The air in the tube undergoes harmonic axial displacement $u(x,t)$ with frequency ω. Do the following:

(1) write the boundary conditions at the two ends of the tube in terms of displacement $u(x,t)$

(2) recall the harmonic solution of the wave equation consisting of the multiplication of two terms:

- **(a)** first term: a function $U(x)$ consisting of trigonometric functions depending on wavenumber γ, space variable x, and constants C_1, C_2
- **(b)** second term: a complex exponential depending on frequency ω and time variable t

(3) substitute (2) into (1) and write the boundary conditions at the two ends of the tube in terms of unknown constants C_1, C_2

(4) solve (3) for C_2 and obtain an equation for C_1

(5) discuss the trivial solution

(6) choose to search for the nontrivial solution and define the equation that needs to be solved to find a nontrivial solution

(7) introduce the notation $z = \gamma L$ into (6) and write the eigenvalue equation in terms of z

(8) solve the eigenvalue equation and write the general expression for the eigenvalues z_j, $j = 1, 2, 3$,...

(9) use (7) to write the general expression for the wavenumber γ in terms of j and length L

(10) Recall from HW01 the definition of wavenumber γ in terms of frequency ω and wavespeed c

(11) solve (10) for frequency ω in terms of wavenumber γ and wavespeed c

(12) use (9) and (11) to write the general expression of the natural frequency ω_j in terms of j, wavespeed c, and tube length L

(13) use (12) to write the general expression of the Hz frequency f in terms of j, wavespeed c, and tube length L

(14) explain the meaning of fundamental frequency and overtones

(15) explain the meaning of fundamental harmonic and higher harmonics

(16) recall the relationship between wavelength λ and wavenumber γ

(17) use (9) into (16) to write the general expression for the wavelength λ_j in terms of j and tube length L

(18) use (17) to explain which wavelength multiples could accommodate vibratory standing waves in a tube of length L closed at both ends

(19) recall $U(x)$ from (2)(a) and use (4) to eliminate C_2

(20) set $C_1 = 1\text{in}$ (19) and write the general expression of the displacement modeshape $U_j(x)$ in terms of the wavenumber γ_j

(21) write the general expression of the pressure modeshapes

B.4 Calculate Air Vibration In A Tube Closed At Both

ENDS Consider a tube of length $L = 0.85\,\mathrm{m}$ closed at both ends. Take sound speed $c = 340\,\mathrm{m/s}$ and air density $\rho = 1.2\,\mathrm{kg/m^3}$. Use the results of Problem B.3 to calculate the modal vibration characteristics and the modeshapes for $j = 1, \ldots, N$ modes with $N = 5$.

(1) Display input data

(2) Calculate and display the modal vibration characteristics:

- **(a)** eigenvalues z_j
- **(b)** wavenumbers γ_j
- **(c)** rad/s frequencies ω_j
- **(d)** Hz frequencies f_j

(3) Calculate and plot the displacement modeshapes

(4) Discuss your results

(5) Graduate work: Calculate and plot the normalized pressure modeshapes and discuss your result in comparison with item (3)

B.5 Calculate The Slide Whistle

Consider the slide whistle shown in Figure 4. The fully retracted whistle is shown in Figure 4a, whereas the fully extended whistle is shown in Figure 4b. The length of the residual cavity of the fully retracted whistle tube can be approximated to be $L_0 = 6\,\mathrm{mm}$ (Figure 4a). The thickness of the piston inside the whistle can be approximated to be $L_p = 6\,\mathrm{mm}$. The thickness of the end walls can be approximated to be $L_w = 6\,\mathrm{mm}$. Assume sound speed in air to be $c = 340\,\mathrm{m/s}$. The purpose of this problem is to calculate the fundamental frequency for various positions of the whistle slider

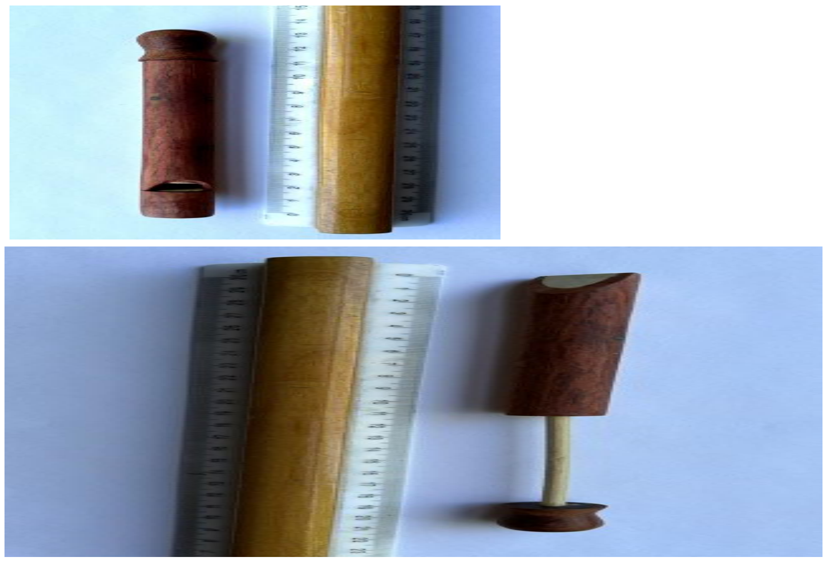

Figure 4 sliding whistle: (a) fully retracted, showing the piston end; (b) fully extended. The ruler major markings are in cm.

Recall the theoretical results of Problem B.3 and do the following:

(1) display input data

(2) use Figure 4 to estimate the fully extended whistle length L_e

(3) use Figure 4 to estimate the fully contracted whistle length L_c

(4) use L_e, L_c to calculate the extension travel dL_I corresponding to fully extended slide whistle

(5) use dL_I and L_0 to calculate the cavity length L_I of a fully extended slide whistle

(6) estimate the fully extended frequency f_I in Hz and find the nearest musical note expressed as a letter name and an octave number (e.g., G3 signifies note G in the 3rd octave)

(7) use L_e, L_c to calculate the extension travel dL_{II} corresponding to 40% extended slide whistle

(8) use dL_{II} and L_0 to calculate the cavity length L_{II} of a 40%- extended slide whistle

(9) calculate the 40%-extended frequency f_{II} in Hz and find the nearest musical note expressed as a letter name and an octave number

(10) comment on your work Graduate work: Estimate the position of the piston to obtain the musical note B7. Do the following:

(11) display the frequency f_{III} of the note B7

(12) calculate the cavity length L_{III} needed to generate frequency f_{III} of the note B7

(13) calculate the extension travel dL_{III} needed to generate frequency f_{III} of the note B7

(14) estimate what percentage of the piston has to be extended to generate the note B7

(15) comment on your work

B.6 Calculate Sound Intensity, Db Scale

Take sound speed $c = 340\,\mathrm{m/s}$ and air density $\rho = 1.2\,\mathrm{kg/m^3}$. Do the following:

(1) Display input data

(2) Calculate air's acoustic impedance Z

(3) Calculate the sound intensity I corresponding to a sound pressure $p = 55$ mPa

(4) Give the values of the reference sound intensity I_0 and reference sound pressure p_0

(5) Covert to dB the sound pressure and the sound intensity of item (3) above

(6) Verify that the sound pressure and sound intensity calculated at item (5) have the same dB value and explain why it is so

(7) Calculate in physical units the sound intensity, sound pressure, and sound power from a loudspeaker with diameter $D = 1.55\,\mathrm{m}$ working at 78 dB

(8) Calculate the sound pressure p_{horn} created by a 118 dB car horn

(9) Give the values of the pain sound intensity I_{pain} and pain sound pressure p_{pain} and comment on the results of items (7) and (8) relative to pain values

(10) Graduate Work: explain why it is necessary to use different dB conversion formulae for sound intensity I and sound pressure p.

3 Extra Credit Challenge Problems

Challenge problems are optional.

C.1 Analyze Air Vibration In A Closed-Open Tube

(1) sketch a tube closed at one end and open at the other end indicating all the information needed for the analysis

(2) write the boundary conditions at the two ends of the tube in terms of displacement $u(x,t)$ and pressure disturbance $p(x,t)$

(3) recall from HW01 the harmonic solution of the wave equation consisting of the multiplication of two terms:

- **(a)** first term: a function $U(x)$ consisting of trigonometric functions depending on wavenumber γ, space variable x, and constants C_1, C_2
- **(b)** second term: a complex exponential depending on frequency ω and time variable t

(4) recall from HW01 the expression of pressure $p(x,t)$ in terms of displacement $u(x,t)$ and bulk modulus K

(5) substitute (3) into (4) to obtain the pressure $p(x,t)$ in terms of constants C_1, C_2

(6) substitute (3) and (5) into (2) and write the boundary conditions at the two ends of the tube in terms of unknown constants C_1, C_2

(7) solve (6) for C_2 and obtain an equation for C_1

(8) discuss the trivial solution

(9) choose to search for the nontrivial solution and define the equation that needs to be solved to find a nontrivial solution

(10) introduce the notation $z = \gamma L$ into (9) and write the eigenvalue equation in terms of z

(11) solve the eigenvalue equation and write the general expression for the eigenvalues z_j, $j = 1, 2, 3$,...

(12) use (10) to write the general expression for the wavenumber γ_j in terms of j and length L

(13) Recall from HW01 the definition of wavenumber γ in terms of frequency ω and wavespeed c

(14) solve (13) for frequency ω in terms of wavenumber γ and wavespeed c

(15) use (12) and (14) to write the general expression of the natural frequency ω_j in terms of j, wavespeed c, and tube length L

(16) use (15) to write the general expression of the Hz frequency f_j in terms of j, wavespeed c, and tube length L

(17) explain the meaning of fundamental frequency and overtones

(18) explain the meaning of fundamental harmonic and higher harmonics

(19) recall from HW01 the relationship between wavelength λ and wavenumber γ

(20) use (12) into (19) to write the general expression for the wavelength λ_j in terms of j and tube length L

(21) use (20) to explain which wavelength multiples could accommodate vibratory standing waves in a tube of length L closed at one end and open at the other end

(22) recall $U(x)$ from (3)(a) and use (7) to eliminate C_2

(23) set $C_1 = 1\text{in}$ (22) and write the general expression of the modeshape $U_j(x)$ in terms of the wavenumber γ_j

(24) plot the normalized displacement modeshapes for the modes number 1 through 5 of air vibration in a tube of length L; discuss your results

(25) Graduate work: calculate and plot the normalized pressure modeshapes for the modes number 1 through 5 of air vibration in a tube of length L; discuss your results

C.2 Analyze Air Vibration In An Open-Open Tube

(1) sketch a tube open at both ends indicating the information needed for analysis

(2) write the boundary conditions at the two ends of the tube

(3) repeat all steps of Problem C.1 paying particular attention to the values taken by C_1 and C_2

C.3 Calculate Air Vibration In A Tube With Various Boundary Conditions

Consider a tube of length $L = 0.85\,\mathrm{m}$. Take sound speed $c = 340\,\mathrm{m/s}$ and air density $\rho = 1.2\,\mathrm{kg/m^3}$. The purpose is to calculate the modal vibration characteristics and the modeshapes for $j = 1, \ldots, N$ modes with $N = 5$. The following boundary conditions will be considered:

boundary conditions case $x = 0$ $x = L$

- **(i)** closed open
- **(ii)** open open

(1) Display input data Then, for each of the two boundary conditions cases, do the following:

(2) Calculate and display the modal vibration characteristics:

- **(a)** eigenvalues z_j
- **(b)** wavenumbers γ_j
- **(c)** rad/s frequencies ω_j
- **(d)** Hz frequencies f_j

(3) Calculate and plot the displacement modeshapes

(4) Discuss your results in comparison with Problem B.4

(5) Graduate work: Calculate and plot the normalized pressure modeshapes and discuss your result in comparison with item (3)